Applied Statistics

Lecture 9:
Cross-Validation vs. Model-based Estimates for Prediction Error

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Learning Objectives for this lecture

- Reflect upon how to estimate the accuracy of predictions.
- Explain the difference between cross-validation and the model-based estimators for prediction error.
- Explain the concepts of apparent error, overall prediction error, and covariance penalty.
- Compare the differences between mallows C_p, SURE, and Akaike information criterion.
- Apply the the model-based estimators for predicition error
- Explain when Akaike information criterion are accurate and apply it for model selection.

Outline of lecture

Introduction

Prediction Problem

Cross-Validation

Covariance penalties

Final remarks

Introduction

Questions:

- How to construct an effective prediction rule
- How to estimate the accuracy of the predictions

In this lecture we concentrate on the latter question.

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Prediction problem

Definition

Let a training set D consisting of N pairs be

$$D = \{(\mathbf{x}_i, y_i), i = 1, 2, ..., N\}, \tag{1}$$

where $\mathbf{x}_i \in \mathcal{X}$ are **predictors** and $y_i \in \mathbb{R}$ are **responses**. Prediction problem refers to finding the **prediction rule** $r_D(\mathbf{x})$ such that

$$\hat{y} = r_D(\mathbf{x}) \tag{2}$$

is the response for any point $\mathbf{x}$ in the sample space $\mathcal{X}$.

Example

Regression: $y_0 \leftarrow$ `prog`.

Example

Table 7.2 *First 7 of $n = 442$ patients in the diabetes study; we wish to predict disease progression at one year "prog" from the 10 baseline measurements* **`age`**, **`sex`**, ..., **`glu`**.

age	sex	bmi	map	tc	ldl	hdl	tch	ltg	glu	prog
59	1	32.1	101	157	93.2	38	4	2.11	87	151
48	0	21.6	87	183	103.2	70	3	1.69	69	75
72	1	30.5	93	156	93.6	41	4	2.03	85	141
24	0	25.3	84	198	131.4	40	5	2.12	89	206
50	0	23.0	101	192	125.4	52	4	1.86	80	135
23	0	22.6	89	139	64.8	61	2	1.82	68	97
36	1	22.0	90	160	99.6	50	3	1.72	82	138
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮

Example

Classification: y_0 is a vector of binary values.

Example

Table 8.2 *Cell infusion data; human cell colonies infused with mouse nuclei in five ratios over 1 to 5 days and observed to see whether they did or did not thrive. Green numbers are estimates $\hat{\pi}_{ij}$ from the logistic regression model. For example, 5 of 31 colonies in the lowest ratio/days category thrived, with observed proportion $5/31 = 0.16$, and logistic regression estimate $\hat{\pi}_{11} = 0.11$.*

Ratio	Time 1	2	3	4	5
1	5/31 **.11**	3/28 **.25**	20/45 **.42**	24/47 **.54**	29/35 **.75**
2	15/77 **.24**	36/78 **.45**	43/71 **.64**	56/71 **.74**	66/74 **.88**
3	48/126 **.38**	68/116 **.62**	145/171 **.77**	98/119 **.85**	114/129 **.93**
4	29/92 **.32**	35/52 **.56**	57/85 **.73**	38/50 **.81**	72/77 **.92**
5	11/53 **.18**	20/52 **.37**	20/48 **.55**	40/55 **.67**	52/61 **.84**

Prediction error quantification

- To quantify the prediction error, one needs to specify the **discrepancy** $d(y, \hat{y})$ between a prediction $\hat{y}$ and the actual response y.
- The two most common choices are **squared error**

$$d(y, \hat{y}) = (y - \hat{y})^2, \tag{3}$$

and the **classification error**

$$d(y, \hat{y}) = \begin{cases} 1 & \text{if } y \neq \hat{y} \\ 0 & \text{if } y = \hat{y} \end{cases} \tag{4}$$

for discrete (dichotomous) values.

True error rate

Definition

Let the z_i be a $p+1$-dimensional vector representing the realisation of the observation pair $(\mathbf{x}_i, y_i)$ in the training set D, corresponding to the random variables $\mathbf{X}_i, Y_i$, be *i.i.d.* observations from the probability distribution F, that is, $\mathbf{Z}_i \equiv (\mathbf{X}_i, Y_i) \sim F$, for $i = 1, ..., N$. The **true error rate** $E_{\text{true},d}$ of the rule r_D is the

$$E_{\text{true},d} = \mathrm{E}_F \left\{ d(Y_0, \hat{Y}_0) \right\}. \tag{5}$$

where $\hat{Y}_0 = r_d(X_0)$, given an independent, new observation pair $(\mathbf{X}_0, Y_0) = \mathbf{Z}_0 \sim F$.

Note: D is held fixed in the expectation, only $(\mathbf{X}_0, Y_0)$ are random.

Supernova Example

Example

- Goal: Create regression from 10 spectral measurements to absolute magnitudes
- Reasonable model $y_i \sim N(\mu_i, 1)$ for all i.
- Estimate by Lasso so that

$$\hat{y} = \mathbf{x}_0^{\mathrm{T}} \tilde{\beta}(\lambda) \qquad (6)$$

- This yielded the mean square error

$$E_{\mathrm{mse}} = \frac{1}{N} \sum_{i=1}^{N} (y_i - \hat{y}_i)^2 = 0.720 \quad (7)$$

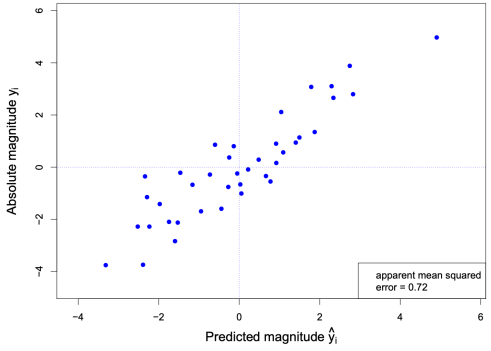


Figure 12.1 The supernova data; observed absolute magnitudes y_i (on log scale) plotted versus predictions $\hat{y}_i$ obtained from lasso rule (12.10), for $N = 39$ nearby Type Ia supernovas. Predictions based on 10 spectral power measurements, 7 of which had nonzero coefficients in $\tilde{\beta}(\lambda)$.

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Cross-validation

- The true error rate E_{true} for the prediction rule r_D is not available.
- An estimate of the error is the average **training error**

$$E_{\text{train}} = \frac{1}{N} \sum_{i=1}^{N} d(y_i, \hat{y}_i). \tag{8}$$

- The error E_{rms} usually underestimates E_{true} (why?)
- An additional independent **validation set**

$$D_{\text{val}} = \{(\mathbf{x}_{0j}, y_{0j}), \quad j = 1, 2, \ldots, N_{\text{val}}\} \tag{9}$$

yields **unbiassed estimate** for E_{true} as

$$E_{\text{val}} = \frac{1}{N_{\text{val}}} \sum_{j} D(y_{0i}, \hat{y}_{0i}). \tag{10}$$

Leave-One-Out Cross-Validation

- However, there is always not an access to large, independent validation set
- **Cross-validation** attempts to mimic E_{val} without the need for a large validation set.

Definition

Let $D(i)$ be the reduced training set in which the pair $(\mathbf{x}_i, y_i)$ has been omitted, and let $r_{D(i)}$ indicate the rule constructed on the basis of $D(i)$. The **leave-one-out cross-validation** estimate of prediction error is

$$E_{\text{cv}} = \frac{1}{N} \sum_{i=1}^{N} d(y_i, \hat{y}_{(i)}), \quad \hat{y}_{(i)} = r_{D(i)}(\mathbf{x}_i). \tag{11}$$

- One can alternatively leave out several pairs at a time

Supernova Example

Example

- Goal: cross-validation estimate using $J = 13$ groups of three cases each.
- The cross-validation error

$$E_{cv} = 1.17 \qquad (12)$$

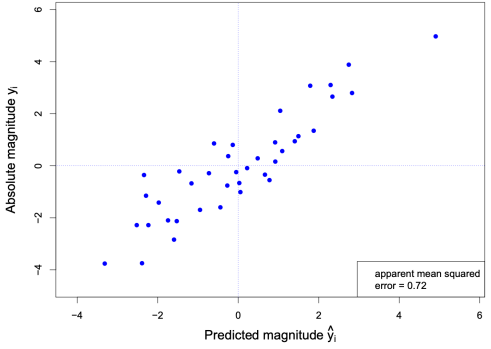


Figure 12.1 The supernova data; observed absolute magnitudes y_i (on log scale) plotted versus predictions $\hat{y}_i$ obtained from lasso rule (12.10), for $N = 39$ nearby Type Ia supernovas. Predictions based on 10 spectral power measurements, 7 of which had nonzero coefficients in $\tilde{\beta}(\lambda)$.

Nested-Cross-Validation

Example

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A method to determine the mammographic regions that show early changes due to the development of breast cancer

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Abstract

It is well understood nowadays that changes in the mammographic parenchymal pattern are an indicator of a risk of breast cancer and we have developed a statistical method that estimates the mammogram regions where the parenchymal changes, due to breast cancer, occur. This region of interest is computed from a score map by utilising the anatomical breast coordinate system developed in our previous work. The method also makes an automatic scale selection to avoid overfitting while the region estimates are computed by a nested cross-validation scheme. In this way, it is possible to recover

Nested-Cross-Validation (2)

Example

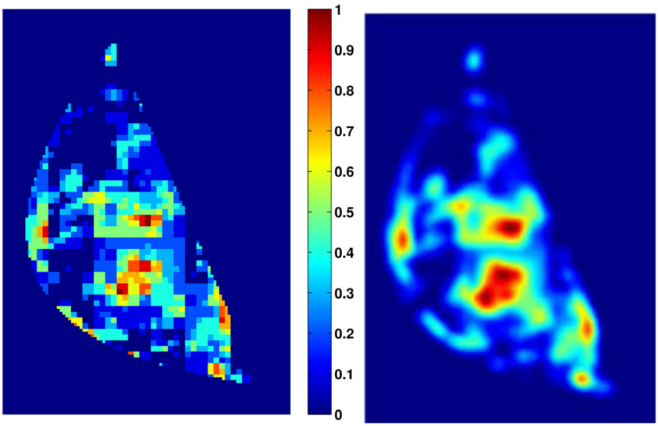


Figure 7. (*a*) The estimated weight map from the Nijmegen study. The colour indicates the percentage of the number of times the respective area was selected to be relevant in the cross-validation scheme. (*b*) The same as (*a*), but the weight map is made smooth with a Gaussian filter to facilitate visual inspection.

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Covariance penalties for Prediction Error Estimation

- Cross-validation is non-parametric
  - No probabilistic modelling needed.
- **Covariance penalties**, for prediction error estimation, are parametric
  - Require probability models.
  - Less noisy than CV
- We will look at
  - **Mallows'** C_p
  - **Stein's unbiased risk estimate** (SURE)
  - **Akaike information criterion**

Covariance penalties: the setting

The covariance penalty approach treats prediction error estimation in a regression framework.

- Assume the predictor vectors x_i in the training set D are fixed (not generated by a random variable).
- Assume that the **unknown** vector μ of expectations, $\mu_i = \mathrm{E}[Y_i]$, has yielded realisation vector of responses y according to some given probability model.
- Assume further that $\mathrm{cov}\{Y\} = \sigma^2 I$, where the variance is **known** (what does this form of covariance mean?)
- Define a regression rule to compute the vector estimate $\hat{\mu} = r(y)$.

Example

The linear regression model $\mu = \mathbf{X}\beta$ yields the LS estimate that can be used as the regression rule as

$$\hat{\mu} = r(\mathbf{y}) = \mathbf{X}(\mathbf{X}^{\mathrm{T}}\mathbf{X})^{-1}\mathbf{X}^{\mathrm{T}}\mathbf{y}, \tag{13}$$

where $\mathbf{X}$ is the $N \times p$ matrix of observations $\mathbf{x}_i^{\mathrm{T}}$, $i, = 1, 2, \ldots, N$ as columns.

Covariance penalties: evaluation of prediction accuracy

- Assuming a new vector of observations $\mathbf{Y}_0$ from the model (mean at μ, uncorrelated and known variance)
- The **overall prediction error** is the expected mean square error

$$E_{\text{overall}} = \frac{1}{N}\sum_i \underbrace{\mathrm{E}_{\mathbf{Y}_0|\hat{\mu}}\left\{(Y_{0i}-\hat{\mu}_i)^2\right\}}_{=:\epsilon^2_{\text{overall},i}} = \frac{1}{N}\mathrm{E}_{\mathbf{Y}_0|\hat{\mu}}\left\{(\mathbf{Y}_0-\mu)^{\mathrm{T}}(\mathbf{Y}_0-\mu)\right\}. \tag{14}$$

- The **mean apparent error** is given by

$$E = \frac{1}{N}\sum_i \underbrace{(y_{0i}-\hat{\mu}_i)^2}_{=:\hat{\epsilon}^2_i} = \frac{1}{N}(\mathbf{y}_0-\mu)^{\mathrm{T}}(\mathbf{y}_0-\mu). \tag{15}$$

Covariance penalties

The two errors are related by the following general lemma:

Lemma

Let $\mathbf{Y}$ *and* $\mathbf{Y}_0$ *independent random vectors where* $E\{\mathbf{Y}\} = E\{\mathbf{Y}_0\} = \mu$ *and* $\mathrm{cov}\{\mathbf{Y}\} = \mathrm{cov}\{\mathbf{Y}_0\} = \sigma^2\mathbf{I}$ *and the variance* σ^2 *is known. Further, let* $\hat{\mu} = r(\mathbf{Y})$*. Then*

$$\mathrm{E}\left\{(Y_{0i} - \hat{\mu}_i)^2\right\} = \mathrm{E}\left\{(Y_i - \hat{\mu}_i)^2\right\} + 2\,\mathrm{cov}(\hat{\mu}_i, y_i). \tag{16}$$

Note: $\mathrm{cov}(\hat{\mu}_i, y_i)$ measures the amount by which y_i influences its own prediction $\hat{\mu}_i$.

- The lemma states that the apparent error underestimates the true prediction error by the **covariance penalty** (the second term on the right).

Covariance penalties: Estimate for the overall prediction error

Theorem

The overall prediction error can be estimated by

$$\hat{E}_{\text{overall}} = E + \frac{2}{N}\sum_{i=1}^{N} \widehat{\text{cov}}(\hat{\mu}_i, y_i). \tag{17}$$

- The form of the term $\widehat{\text{cov}}(\hat{\mu}_i, y_i)$ depends on context of the particular the prediction problem.

Covariance penalties: Linear prediction model

Theorem

Let $\mathbf{Y}$ *be a random vector and given the prediction model* $\hat{\mu} = \mathbf{c} + \mathbf{M}\mathbf{Y}$ *for some known* N*-vector* c *and* $N \times N$ *matrix* M*. The covariance matrix between* $\hat{\mu}$ *and* $\mathbf{Y}$ *is*

$$\operatorname{cov}(\hat{\mu}, \mathbf{Y}) = \sigma^2 \mathbf{M} \tag{18}$$

that yields the overall prediction error estimate

$$\hat{E}_{\text{overall}} = \frac{1}{N} \sum_{i=1}^{N} (y_i - \hat{\mu}_i)^2 + \frac{2\sigma^2}{N} \operatorname{trace}(\mathbf{M}). \tag{19}$$

This formula is called **Mallows'** C_p estimate of prediction error.

Covariance penalties: OLS Regression

Corollary

For OLS estimation, $\mathbf{M} = \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T$*, hence,* $\mathrm{trace}(\mathbf{M}) = p$*, where* p *is the number of predictors. This now yields the overall prediction error estimate*

$$\hat{E}_{\text{overall}} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{\mu}_i)^2 + \frac{2\sigma^2 p}{N}. \tag{20}$$

Comparing the equation above with the equation of the covariance penalty earlier yields a general definition of degrees of freedom for a regression rule.

Definition

$$\text{dof} = \frac{1}{\sigma^2}\sum_{i=1}^{N}\widehat{cov}(\hat{\mu}_i, y_i) \tag{21}$$

This definition grounds comparison of different types of regression models (how?).

Covariance penalties: Lasso

- For Lasso problems, setting p to the number of nonzero reggression coefficients yields a good approximation
- This allows us to use the previous OLS covariance penalty formula for the overall prediction error.

Covariance penalties: SURE

Theorem

Let $\mathbf{Y} \sim N(\mu, \sigma^2\mathbf{I})$ be a normally distributed random vector. Then, for any differentiable estimator $\hat{\mu} = r(Y)$, the covariance matrix between $\hat{\mu}$ and $\mathbf{Y}$ is

$$\operatorname{cov}(\hat{\mu}, \mathbf{Y}) = \sigma^2 \mathrm{E}\left\{\frac{\partial \hat{\mu}_i}{\partial y_i}\right\} \tag{22}$$

that yields the overall prediction error estimate

$$\hat{E}_{\text{overall}} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{\mu}_i)^2 + \frac{2\sigma^2}{N}\sum_{i=1}^{N}\frac{\partial \hat{\mu}_i}{\partial y_i}. \tag{23}$$

*This is known as Stein's unbiased risk estimator (**SURE**).*

Note: here we do not need the assumption of linearity.

Covariance penalties: SURE (2)

Example

For the Kidney fitness data, $\widehat{\text{dof}} = \sum_{i=1}^{N} \frac{\partial \hat{\mu}_i}{\partial y_i} = 6.67$.

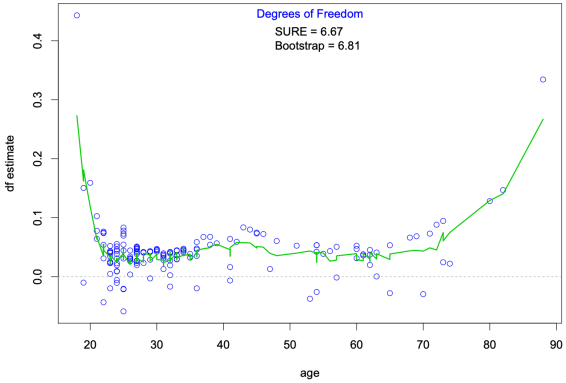


Figure 12.4 Analysis of the `lowess(x,y,1/3)` fit to kidney data of Figure 1.2. Open circles are SURE coordinate-wise df estimates $\partial \hat{\mu}_i / \partial y_i$, plotted versus age_i, giving total degrees of freedom 6.67. The solid curve tracks bootstrap coordinate-wise estimates (12.65), with their sum giving total df $= 6.81$.

Covariance Penalties: Parametric Bootstrap

Parametric bootstrap estimate for the covariance

Assume that $\mathbf{y}$ is a realisation generated by a member $f_\mu(\mathbf{y})$ in a parametric family

$$\mathcal{F} = \{f_\mu(\mathbf{y}), \mu \in \Omega\} \tag{24}$$

that yields $\hat{\mu} = r(\mathbf{y})$. Hence,

$$f_\mu(\mathbf{y}) \to \mathbf{y} \to \hat{\mu} = r(y). \tag{25}$$

The bootstrap sample drawn in analogy

$$f_\mu(\mathbf{y}^*) \to \mathbf{y}^* \to \hat{\mu}^* = r(y^*) \tag{26}$$

that yield the bootstrap estimate for the covariance matrix between $\hat{\mu}$ and $\mathbf{Y}$

$$\text{cov}(\hat{\mu}_i, y_i) = \frac{1}{B}\sum_{b=1}^{B}(\hat{\mu}_i^{*b} - \hat{\mu}_i)(y_i^{*b} - \hat{y}_i). \tag{27}$$

Covariance Penalties: Classification

Example

Consider classification problem where the observation y_i and predictor μ_i are binary, with the error

$$d(y_i, \hat{\mu}_i) = \begin{cases} 1 & \text{if } y_i \neq \hat{\mu}_i \\ 0 & \text{if } y_i = \hat{\mu}_i. \end{cases} \tag{28}$$

This choice leads to the overall prediction estimate

$$\hat{E}_{\text{overall}} = \frac{\#\{y_i \neq \hat{\mu}_i\}}{N} + \frac{2}{N} \sum_{i=1}^{N} \text{cov}(\hat{\mu}_i, y_i), \tag{29}$$

where

$$\text{cov}(\hat{\mu}_i, y_i) = P(y_i = 1)\,(1 - P(y_i = 1))\,(P(\hat{\mu}_i = 1 | y_i = 1) - P(\hat{\mu}_i = 1 | y_i = 0)). \tag{30}$$

Covariance Penalties: Akaike Information Criterion

Example

Let obervations y_i are obtained from different memebers of a one-parameter exponential family $f_\mu(y)$.
Error is measured by the deviance based on **Kullback–Leibler divergence**

$$d(y, \hat{\mu}) = 2d_{\mathrm{KL}}(f_y||f_{\hat{\mu}}) = 2\int f_y(z)\log\left(\frac{f_y(z)}{f_{\hat{\mu}(z)}}\right)dz. \tag{31}$$

Assuming generalized linear model with p degree of freedom, the covariance penalty takes the form $\frac{2p}{N}$. The overall prediction error hence is

$$\hat{E}_{\text{overall}} = -\frac{2}{N}\underbrace{[\log f_{\hat{\mu}}(y) - p]}_{\text{Akaike Information Criterion (AIC)}} + \text{const.} \tag{32}$$

Model selection by AIC: select model that maximises the penalised log-likelihood $\log f_{\hat{\mu}}(y) - p$.

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Introduction

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Final remarks

- Covariance penalty estimates should be preferred to cross-validation when believable parametric models are available.
- The distance between train and validation sets is not irrelevant.
- Validation error estimates are more believable if the test set is further separated from the training set.
- Prediction over large time interval is challenging if validation sets cannot set long apart from the training intervals.

Training–Validation Example

Example

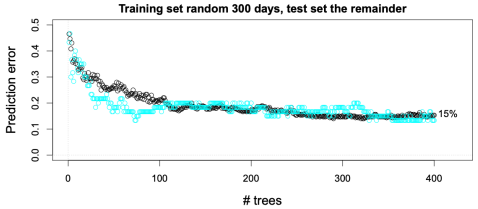


Figure 12.6 Test error (blue) and cross-validated training error (black), for Random Forest prediction rules using the imaginary medical study (12.81)–(12.82). *Top panel*: training set randomly selected 300 days, test set the remaining 60 days. *Bottom panel*: training set the first 300 days, test set the last 60 days.